

Problem 8

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Problem 1 (a) Let f be a positive differentiable function, defined on an open interval I . Find the formula for the derivative of the function $\ln(f(x))$.

Explanation. $\frac{d}{dx} \ln(f(x)) = \frac{1}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)}$

(b) Using the formula obtained in part (a), compute the derivatives of the following functions.

(i) $f(x) = \ln(x^2 + x + 1)$

Explanation. $f'(x) = \frac{2x + 1}{x^2 + x + 1}$

(ii) $f(x) = \ln(\sec(x) + \tan(x))$

Explanation. $f'(x) = \frac{\sec(x)\tan(x) + \sec^2(x)}{\sec(x) + \tan(x)} = \frac{\sec(x)(\tan(x) + \sec(x))}{\sec(x) + \tan(x)} = \sec(x)$

(iii) $f(x) = \ln(\ln(x))$

Explanation. $f'(x) = \frac{\frac{1}{x}}{\ln(x)} = \frac{1}{x \ln(x)}$